SABD Project: Urban Mobility Analysis

Part 2 - Data Handling with R and GIS

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Table of Contents

1 Statement **2**
1.1 Objectives of the Part 2 . . . 2
1.2 Loading the data . . . 2
1.3 Part Two - Data processing . . . 2
1.4 Vocabulary . . . 2
1.5 Checking the structure of the database . . . 2
1.6 New Variables Definition . . . 5
1.7 Reduction to the trip level . . . 8
1.8 Reduction to the person level . . . 9
1.9 Reduction to the household level . . . 9
1.10 External calculus . . . 10
1.11 Created variables . . . 11
1.12 Saving the three data frames . . . 11
1.13 **Your** data processing . . . 12

2 GIS Part **13**

3 Map and Calculus of area indicators: CARTE NBD BTT **14**

4 Colored mapping **15**

5 Map with labels **15**

6 Complete map of daily travel times **16**

7 References **19**
7.1 Sources . . . 19

8 Appendix **20**

1 Statement

1.1 Objectives of the Part 2

Part 1 gave the framework of the project (topics, unit of analysis, evaluation protocol). Part 2 guides you through the **data handling**:

- load the survey data and **check the structure** (one row per route, immobile persons, identifiers);
- **create and modify variables** with R, and with MongoDB;
- reduce the database to the **trip**, **person** and **household** levels (`allgreD`, `allgreI`, `allgreM`);
- compute **indicators by area** and draw **maps** of the Grenoble area (average daily travel time, number of trips, share of immobile persons).

You will then continue with the data handling for **your own topic**.

1.2 Loading the data

First lets load the data

```
load("allgre.PB_V26.RData")
load("allgre_3DF.RData")
```

1.3 Part Two - Data processing

This part reduces the database to the **trip**, **person** and **household** levels, creates new variables, and explains how to aggregate information from a lower level to a higher one. Scripts use `tidyverse` functions only.

1.4 Vocabulary

Term	Identifier	Meaning
Route	`id_traj` (`NO_TRAJ`)	One segment of a trip, with one mode. **One row of the database = one route.**
Trip	`id_depl` (`NO_DEPL`)	Movement between two activities, made of one or several routes
Person	`id_pers` (`NO_PERS`)	Surveyed member of a household
Household	`id_men`	Group of persons

1.5 Checking the structure of the database

Rows are routes: variables of a person (such as `nbd`, the number of trips of the person) are **repeated on each row**. To count persons or trips, never count rows.

```
tibble(rows       = nrow(allgre.PB_V26),
       households = n_distinct(allgre.PB_V26$id_men),
       persons    = n_distinct(allgre.PB_V26$id_pers),
       routes     = n_distinct(allgre.PB_V26$id_traj))
```

```
## # A tibble: 1 x 4
##    rows households persons routes
##   <int>      <int>   <int>  <int>
## 1 34290       3802    8861  25273
```

```
# Distribution of the number of trips PER PERSON (one value per person)
allgre.PB_V26 %>% distinct(id_pers, nbd) %>% count(nbd) #%>% print(n = 30)
```

```
##    nbd    n
## 1    0 1436
## 2    1   93
## 3    2 2108
## 4    3  734
## 5    4 1953
## 6    5  761
## 7    6  766
## 8    7  358
## 9    8  264
## 10   9  159
## 11  10   87
## 12  11   54
## 13  12   41
## 14  13   20
## 15  14   11
## 16  15    7
## 17  16    5
## 18  17    1
## 19  18    1
## 20  22    2
```

1.5.1 Number of trips

Analyzing the number of trips, you should check that 44 observations are: `nbd = 22`

These are persons that declare 22 trips in their day. For one individual this value is repeated 22 times in the DF ! We deduce that only 2 persons have done 22 trips in their day...

```
table(allgre.PB_V26$dispovp, useNA = "ifany")
```

```
##
##     1     2     3  <NA>
## 12884  1683  2988 16735
```

```
names(allgre.PB_V26)
```

```
##  [1] "tir"            "NO_MEN"         "NO_PERS"        "NO_DEPL"
##  [5] "zoneres.x"      "jourdepl"       "TYPE_HAB"       "TYPE_OCU"
##  [9] "Gare2"          "Gare5"          "telefon"        "annuaire"
## [13] "internet"       "VP_DISPO"       "GENRE1"         "ENERGIE1"
## [17] "AN_VP1"         "PUIS_VP1"       "POSSES1"        "LIEU_STAT1"
## [21] "TYPE_STAT1"     "GENRE2"         "ENERGIE2"       "AN_VP2"
## [25] "PUIS_VP2"       "POSSES2"        "LIEU_STAT2"     "TYPE_STAT2"
## [29] "GENRE3"         "ENERGIE3"       "AN_VP3"         "PUIS_VP3"
## [33] "POSSES3"        "LIEU_STAT3"     "TYPE_STAT3"     "GENRE4"
## [37] "ENERGIE4"       "AN_VP4"         "PUIS_VP4"       "POSSES4"
## [41] "LIEU_STAT4"     "TYPE_STAT4"     "NB_velo"        "NB_2Rm"
## [45] "COEF_MNG"       "zoneres.y"      "sexe"           "lien"
## [49] "age"            "TEL_PORT"       "mail"           "permis"
## [53] "etabscol"       "OCCU1"          "OCCU2"          "csp"
## [57] "ABO_TC"         "VAL_ABO"        "travdom"        "zonetrav"
## [61] "dispovp"        "PBM_STAT"       "STAT_TRAV"      "fqvelo"
## [65] "FQ2R1"          "FQ2R2"          "fqvpcond"       "fqvppass"
## [69] "freqtcu"        "freqtram"       "freqrurb"       "freqtransisere"
## [73] "freqter"        "situveil"       "zoneres.x.1"
```

```
##  [ reached 'max' / getOption("max.print") -- omitted 41 entries ]
```

```
allgre.PB_V26 %>% select(dispovp, NO_MEN, NO_PERS, NO_DEPL, tir, VP_DISPO) -> DF
addmargins(table(DF$dispovp, DF$VP_DISPO, useNA = "ifany"))
```

```
##
##            0     1     2     3     4     5   Sum
##   1        2  3067  7715  1618   442    40 12884
##   2        4   904   661   100    14     0  1683
##   3     1095  1121   659   106     7     0  2988
##   <NA>  1482  6153  7485  1311   291    13 16735
##   Sum   2583 11245 16520  3135   754    53 34290
```

```
# or
DF <- allgre.PB_V26 %>% dplyr::select(sexe, nbd) %>% table()
kable(DF)
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	22
677	66	2322	1080	4123	1967	2345	1338	883	727	402	198	156	117	70	15	32	0	18	22
759	52	2393	1264	4070	2003	2367	1219	1265	717	476	396	341	147	84	90	50	17	0	22

`nbd` was determined counting the trips in the DF for each individual. `NO_DEPL` is the number of trips determined by the survey institution.

Studying the number of trips (`NO_DEPL`), we see that 2 trips are numbered 22.

```
# Check of NO_DEPL - number of trips
table(allgre.PB_V26$NO_DEPL, useNA = "ifany")
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
## 1436 8033 7743 5373 4672 2589 1800 1019  658  390  229  142   91   48   27   16
##   16   17   18   19   20   21   22
##    9    4    3    2    2    2    2
```

We can check that each trip (observation) is associated to an unique `id`.

We see that 1436 trips have no number. These are lines for individuals without any trips: immobile persons

A half empty line describe the individual but not the trip

Identification key is created for each trip using individual ID and the trip number

```
# Creation of new id_depl
allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(id_depl = if_else(is.na(NO_DEPL), NA_character_,
                           paste(id_pers, NO_DEPL, sep = "_")))
table(is.na(allgre.PB_V26$id_depl))
```

```
##
## FALSE
## 34290
```

```
table(allgre.PB_V26$nbd)
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
## 1436  118 4715 2344 8193 3970 4712 2557 2148 1444  878  594  497  264  154  105
##   16   17   18   22
```

```
##   82   17   18   44
```

1.6 New Variables Definition

1.6.1 Immobil person

`immobil = 1` if the person made **no** trip (`nbd = 0`), 0 otherwise.

```
allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(immobil = as.integer(nbd == 0))

allgre.PB_V26 %>% distinct(id_pers, .keep_all = TRUE) %>%
  count(immobil, useNA = "ifany")
```

```
##   immobil useNA    n
## 1       0 ifany 7425
## 2       1 ifany 1436
```

1.6.2 Driving licence

`permis` is coded 0 = no, 1 = yes, 2 = accompanied driving (learner, usually under 18). We keep the original variable and add a labelled factor.

```
allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(permis_cat = factor(permis, levels = c(0, 1, 2),
                             labels = c("no licence", "licence", "accompanied driving")))

allgre.PB_V26 %>% distinct(id_pers, .keep_all = TRUE) %>%
  group_by(permis_cat) %>%
  summarise(persons = n(), age_min = min(age, na.rm = TRUE),
            age_median = median(age, na.rm = TRUE), .groups = "drop")
```

```
## # A tibble: 3 x 4
##   permis_cat          persons age_min age_median
##   <fct>                 <int>   <int>      <dbl>
## 1 licence                5896      18         49
## 2 accompanied driving    2330       5         14
## 3 <NA>                    635       1          3
```

> The modality 2 is neither "no licence" nor "licence". For topics using `permis` (1, 2, 6) you must **decide and justify** how it is treated (separate category, merged with 0, or excluded) and mention that it is strongly related to `age`.

1.6.3 Area numbers

Origin and destination areas are recoded by an **integer division** by 1000 (the former `round(zone/1000)` sent the codes from x500 to the next sector).

```
allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(ORIG = zoneorig %/% 1000,
         DEST = zonedest %/% 1000)

head(allgre.PB_V26$ORIG); head(allgre.PB_V26$DEST)
```

```
## [1] 101 314 101 119 101 101
```

```
## [1] 314 101 119 101 101 101
```

```r
table(allgre.PB_V26$ORIG, useNA = "ifany")
```

```
##
##  101  102  103  104  105  106  107  108  109  110  111  112  113  114  115  116
##  913  706  618  338  290  668  223  307  319  485  245  497  478  181  481  499
##  117  118  119  120  121  122  123  124  125  126  127  128  129  130  131  132
##  167  168  626  859  192  806  626  175  236  284  305  585  235  325  514  482
##  133  134  135  136  137  138  139  140  141  142  143  201  202  203  204  205
##  178  396  541  236  286  233  352  283  199  226  314  353  206  272  262  241
##  301  302  303  304  305  306  307  308  309  310  311  312  313  314  401  402
##  203  307  313  233  449  290  264  205  240  318  184  232  291  229  249  339
##  403  501  502  503  504  505  506  507  508  509  510
##  361 1267  179  155  421  273  269  282  159  219  218
##  [ reached 'max' / getOption("max.print") -- omitted 24 entries ]
```

To check in the variables dictionary: the coding of `zoneorig` / `zonedest` and the correspondence with the sector codes (`tir`, which is identical to `zoneres.y`).

1.6.4 Factors and recodings

Qualitative variables are defined as **factors** with explicit labels. Two rules: **list the codes** (`levels =`) instead of relying on the sorted order of the codes present, and **look at the table before and after** each recoding.

```r
# Check the codes actually present, and compare them with the dictionary
cat("OCCU1 codes found:", paste(sort(unique(allgre.PB_V26$OCCU1)), collapse = ", "), "\n")
```

```
## OCCU1 codes found: 1, 2, 3, 4, 5, 6, 7, 8, 9
```

```r
# Codes 1 to 9, in the order of the dictionary (TO CHECK)
status_labels <- c("TRAVAIL A TEMPS PLEIN", "TRAVAIL A TEMPS PARTIEL", "FORMATION STAGE",
                   "ETUDIANT", "SCOLAIRE", "CHOMEUR", "RETRAITE", "RESTE AU FOYER", "AUTRE")

allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(statut_detail = factor(OCCU1, levels = 1:9, labels = status_labels),
         statut2 = fct_collapse(statut_detail,
                                ACTIF = c("TRAVAIL A TEMPS PLEIN", "TRAVAIL A TEMPS PARTIEL",
                                          "FORMATION STAGE")))

table(allgre.PB_V26$statut_detail, useNA = "ifany")
```

```
##
##   TRAVAIL A TEMPS PLEIN TRAVAIL A TEMPS PARTIEL         FORMATION STAGE
##                   12785                    3448                     293
##                ETUDIANT                SCOLAIRE                CHOMEUR
##                    1746                    5820                    1259
##                RETRAITE          RESTE AU FOYER                   AUTRE
##                    6338                    1440                     646
##                    <NA>
##                     515
```

```r
table(allgre.PB_V26$statut2, useNA = "ifany")
```

```
##
##          ACTIF       ETUDIANT       SCOLAIRE        CHOMEUR       RETRAITE
##          16526           1746           5820           1259           6338
## RESTE AU FOYER          AUTRE           <NA>
```

```
##           1440            646            515
```

```
# Before recoding: is the code 9 already used?
table(trunc(allgre.PB_V26$csp / 10), useNA = "ifany")
```

```
##
##    1    2    3    4    5    6    8    9 <NA>
##  462 1502 5765 6188 6820 4000 8738  300  515
```

```
allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(cspgroup = case_when(csp %in% c(81, 82) ~ 7,
                              is.na(csp)         ~ 9,   # unknown (check that 9 is free)
                              TRUE               ~ trunc(csp / 10)),
         cspgroup = factor(cspgroup))

table(allgre.PB_V26$cspgroup, useNA = "ifany")
```

```
##
##    1    2    3    4    5    6    7    8    9
##  462 1502 5765 6188 6820 4000 1172 7566  815
```

The trip purposes (`motifor` at origin, `motifdes` at destination) are reduced to 10 groups by the function below. Immobile persons have no trip, hence **no purpose** (`NA`); a missing purpose for a mobile person goes to "AUTRES".

```
reduce_purpose <- function(x) {
  case_when(
    x %in% c(1, 2)             ~ 1L,   # home & second residence
    x %in% c(11, 12)           ~ 2L,   # work
    x %in% c(21, 22, 26)       ~ 3L,   # primary school
    x %in% c(23, 24, 27, 28)   ~ 4L,   # middle / high school
    x %in% c(25, 29)           ~ 5L,   # university
    x > 30 & x < 44            ~ 6L,   # shopping & services
    x > 49 & x < 55            ~ 7L,   # leisure
    x > 60 & x < 75            ~ 8L,   # escorting
    x == 81                    ~ 10L,  # professional tour
    x == 91 | is.na(x)         ~ 9L,   # other / missing
    TRUE                       ~ NA_integer_   # code not covered: inspect!
  )
}

purpose_labels <- c("DOMICILE", "TRAVAIL", "ECOLE PRIMAIRE", "C.E.S.-LYCEE", "UNIVERSITE",
                    "ACHAT", "LOISIR", "ACCOMPAGNEMENT", "AUTRES", "Professionnel")

allgre.PB_V26 <- allgre.PB_V26 %>%
  mutate(mot_o_red = factor(if_else(nbd == 0, NA_integer_, reduce_purpose(motifor)),
                            levels = 1:10, labels = purpose_labels),
         mot_d_red = factor(if_else(nbd == 0, NA_integer_, reduce_purpose(motifdes)),
                            levels = 1:10, labels = purpose_labels))

table(allgre.PB_V26$mot_o_red, useNA = "always")
```

```
##
##       DOMICILE        TRAVAIL ECOLE PRIMAIRE   C.E.S.-LYCEE     UNIVERSITE
##          12691           4845            914           1173            552
##          ACHAT         LOISIR ACCOMPAGNEMENT         AUTRES  Professionnel
##           4964           4383           3091            185             56
```

```
##           <NA>
##           1436
```

```
# Codes of mobile persons that are NOT covered by the grouping
table(allgre.PB_V26$motifor[is.na(allgre.PB_V26$mot_o_red) & allgre.PB_V26$nbd > 0], useNA = "ifany")
```

```
## < table of extent 0 >
```

1.7 Reduction to the trip level

A trip is made of one or several routes. We count the routes of each trip and keep **one row per trip** (the first route). Persons without trip (immobile) are excluded from this table.

```
allgreD <- allgre.PB_V26 %>%
  filter(!is.na(id_depl)) %>%
  group_by(id_depl) %>%
  mutate(n_routes = max(NO_TRAJ, na.rm = TRUE)) %>%
  ungroup() %>%
  distinct(id_depl, .keep_all = TRUE)

nrow(allgreD)
```

```
## [1] 32843
```

```
table(allgreD$n_routes)
```

```
##
##  -Inf     0     1     2     3     4
##  1436  7404 22799   964   237     3
```

1.7.1 Average distances per area

Distances are averaged on **trips** (not on routes, which would count multi-route trips several times), by residence area.

```
allgreD %>%
  summarise(across(c(D12, D13), list(mean = ~ mean(.x, na.rm = TRUE),
                                     n_na = ~ sum(is.na(.x)))))
```

```
## # A tibble: 1 x 4
##   D12_mean D12_n_na D13_mean D13_n_na
##      <dbl>    <int>    <dbl>    <int>
## 1    4404.     2345     6.27     2345
```

```
dist_by_zone <- allgreD %>%
  group_by(zoneres.y) %>%
  summarise(n_trips = n(),
            across(c(D12, D13), ~ mean(.x, na.rm = TRUE), .names = "{.col}_mean"),
            n_na_D12 = sum(is.na(D12)),
            .groups = "drop")
head(dist_by_zone)
```

```
## # A tibble: 6 x 5
##   zoneres.y n_trips D12_mean D13_mean n_na_D12
##       <int>   <int>    <dbl>    <dbl>    <int>
## 1         1   10597    4068.     5.83      701
## 2         2    9127    4203.     5.97      659
## 3         3    5934    4382.     6.23      385
```

```
## 4         4    1657    4864.     6.92       94
## 5         5    1005    4884.     6.95       90
## 6         6     930    6065.     8.62       57
```

Averages by area are **not stored** in the data frames; the same computation is used in the GIS part.

1.8 Reduction to the person level

Individual indicators computed from trips (sum of durations, number of trips) are calculated **in the trip table**, then joined to the person table. The person table is built from the **complete** base so that **immobile persons are kept**.

```
# Function: names of the columns that do not vary within groups of rows
constant_within <- function(df, id) {
  n_id <- length(unique(df[[id]]))
  names(df)[vapply(names(df),
                   function(v) nrow(unique(df[, c(id, v)])) == n_id,
                   logical(1))]
}

# 1. Indicators computed on trips
trip_agg <- allgreD %>%
  group_by(id_pers) %>%
  summarise(btt_sum    = sum(duree, na.rm = TRUE),    # daily travel time, from declared durations
            n_na_duree = sum(is.na(duree)),           # missing durations
            nbd_chk    = n_distinct(NO_DEPL),         # number of trips counted here
            D12_sum    = sum(D12, na.rm = TRUE),     # daily travel distance
            .groups = "drop")

# 2. Columns of the base that are constant within a person (= person & household variables)
person_cols <- constant_within(allgre.PB_V26, "id_pers")

# 3. One row per person, immobile included
allgreI <- allgre.PB_V26 %>%
  distinct(id_pers, .keep_all = TRUE) %>%
  select(any_of(person_cols)) %>%
  left_join(trip_agg, by = "id_pers") %>%
  mutate(across(c(btt_sum, n_na_duree, nbd_chk), ~ replace_na(.x, 0)),
         # daily travel time with 0 for immobile persons
         btt_all = if_else(immobil == 1, 0, btt))

nrow(allgreI); sum(allgreI$immobil == 1, na.rm = TRUE)
```

```
## [1] 8861
```

```
## [1] 1436
```

Consistency checks. `btt` (from the preprocessing, corrected for errors in declared times) and `btt_sum` (raw sum of declared durations) are expected to differ for some persons: **keep `btt`** for your analyses, and use `btt_sum` only to understand the differences.

1.9 Reduction to the household level

Household size and household licence are computed from the **complete** person table (with immobile persons). The household licence is 1 if **at least one member has a full licence** (`permis == 1`); accompanied driving (2) does not count.

```r
# Is the maximum person number equal to the number of persons ?
allgreI %>% group_by(id_men) %>%
  summarise(n = n(), max_no = max(NO_PERS), .groups = "drop") %>%
  summarise(households_with_gap = sum(n != max_no))
```

```
## # A tibble: 1 x 1
##   households_with_gap
##                 <int>
## 1                   0
```

```r
hh_agg <- allgreI %>%
  group_by(id_men) %>%
  summarise(taillemng  = n(),
            Permis_mng = as.integer(any(permis == 1, na.rm = TRUE)),
            n_licence  = sum(permis == 1, na.rm = TRUE),
            n_mobile   = sum(immobil == 0, na.rm = TRUE),
            .groups = "drop")

# Household variables are added to the person table...
allgreI <- allgreI %>% left_join(hh_agg, by = "id_men")

# ... and the household table keeps only the columns constant within a household
household_cols <- constant_within(allgreI, "id_men")
allgreM <- allgreI %>%
  distinct(id_men, .keep_all = TRUE) %>%
  select(any_of(c(household_cols, "taillemng", "Permis_mng", "n_licence", "n_mobile")))

addmargins(table(allgreM$taillemng))
```

```
##
##    1    2    3    4    5    6    7    8  Sum
## 1124 1379  523  525  206   37    6    2 3802
```

```r
addmargins(table(allgreM$Permis_mng))
```

```
##
##    0    1  Sum
##  308 3494 3802
```

1.10 External calculus

1.10.1 Modal share (per trip)

```r
allgreD %>% count(mode_V2) %>% mutate(share = round(100 * n / sum(n), 1))
```

```
## # A tibble: 6 x 3
##   mode_V2     n share
##   <chr>   <int> <dbl>
## 1 Autre    1691   5.1
## 2 MAP      7404  22.5
## 3 TCIU      546   1.7
## 4 TCU      2284   7
## 5 VP      19482  59.3
## 6 <NA>     1436   4.4
```

1.10.2 Trip duration

```
summary(allgreD$duree)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    1.00    5.00   12.00   18.83   22.00  525.00    1436
```

```
table(is.na(allgreD$duree))
```

```
##
## FALSE  TRUE
## 31407  1436
```

```
table(cut(allgreD$duree, breaks = c(0, seq(30, max(allgreD$duree, na.rm = TRUE) + 30, 30)),
          include.lowest = TRUE), useNA = "always")
```

```
##
##    [0,30]   (30,60]   (60,90]  (90,120] (120,150] (150,180] (180,210] (210,240]
##     27590      2870       566       180        72        32        37        20
## (240,270] (270,300] (300,330] (330,360] (360,390] (390,420] (420,450] (450,480]
##        12         9         4         6         3         5         0         0
## (480,510] (510,540]      <NA>
##         0         1      1436
```

1.11 Created variables

Table 1: Variables created in this part

variable	level	definition
id_depl	trip	Trip key: person id, underscore, trip number (NA for immobile persons)
immobil	person	1 if the person made no trip (nbd = 0), else 0
permis_cat	person	Licence as factor: no licence / licence / accompanied driving
ORIG, DEST	trip	Origin / destination sector: zone code %/% 1000
statut_detail	person	Occupation (OCCU1) with 9 labelled modalities
statut2	person	Occupation grouped (ACTIF = full-time, part-time, training)
cspgroup	person	Socio-professional group (csp / 10; 7 for pcs 81-82; 9 if missing)
mot_o_red	trip	Purpose at origin, 10 groups (NA for immobile persons)
mot_d_red	trip	Purpose at destination, 10 groups (NA for immobile persons)
n_routes	trip	Number of routes of the trip
btt_sum	person	Sum of declared trip durations (raw), 0 if no trip
nbd_chk	person	Number of trips recounted from the trip table
btt_all	person	btt, with 0 for immobile persons
taillemng	household	Number of persons in the household (complete person table)
Permis_mng	household	1 if at least one member has a licence (permis = 1)
n_licence	household	Number of members with licence (permis = 1)
n_mobile	household	Number of mobile members

1.12 Saving the three data frames

The three tables are saved in a single file, used by all the following parts of the project.

```
save(allgreD, allgreI, allgreM, file = "allgre_3DF.RData")
tibble(table = c("allgreD (trips)", "allgreI (persons)", "allgreM (households)"),
       rows  = c(nrow(allgreD), nrow(allgreI), nrow(allgreM)),
       columns = c(ncol(allgreD), ncol(allgreI), ncol(allgreM)))
```

```
## # A tibble: 3 x 3
##   table                rows columns
##   <chr>               <int>   <int>
## 1 allgreD (trips)     32843     126
## 2 allgreI (persons)    8861      93
## 3 allgreM (households) 3802      50
```

1.13 Your data processing

You should start here. Load the three data frames (they were just saved above):

```
load("allgre_3DF.RData")   # allgreD (trips), allgreI (persons), allgreM (households)
```

In this part, we work with geographical data to produce maps
Instructions in chunks transform the data and perform some calculus for GIS maps, they should be reviewed in Mongo
Note that the functions for map creation maybe time and resource consuming ! That is why the chunk are `eval=FALSE`: you will have to turn them on if you want to draw the map.

```
#Parameter to turn TRUE to draw maps (or FALSE)
map_draw <- TRUE
```

2 GIS Part

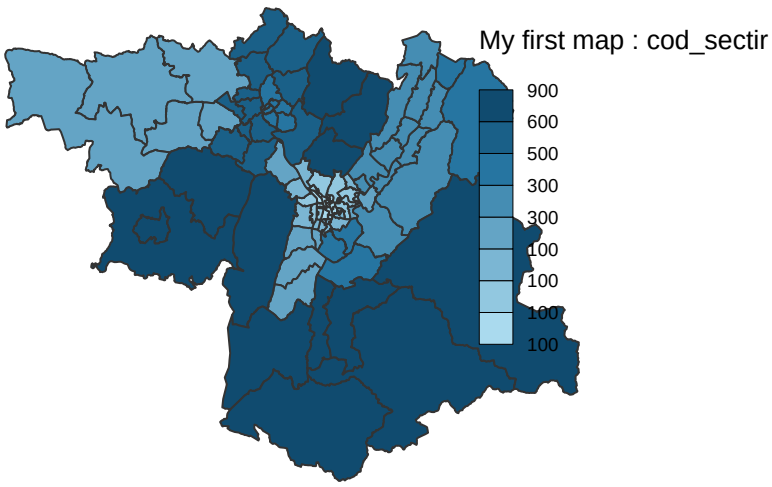
```
# install.packages("sf")
# install.packages("terra")
# install.packages("spData")
# install.packages("spDataLarge", repos = "https://nowosad.r-universe.dev")

library(sf)
library(cartography)
library(terra)
library(spData)        # load geographic data
#library(spDataLarge)
library(RColorBrewer)
newESRI <- st_read("FOND_MAP/newESRI.shp")
```

```
## Reading layer 'NewESRI' from data source
##   'G:\Ira_Lessons_with_Rmd\2021_SABD_SmartAnalytics\2025_TP_EMD\Files_OK_2026\FOND_MAP\NewESRI.shp'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 97 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 4.918609 ymin: 44.6959 xmax: 6.359307 ymax: 45.53372
## Geodetic CRS:  WGS 84
```

```
#st_read("G:/Ira_Lessons_with_Rmd/2021_SABD_SmartAnalytics/2022_TP_EMD/FILES_OK/FOND_MAP/newESRI.shp")

choroLayer(
  x = newESRI, var = "cod_sectir",
  legend.horiz = FALSE,
  legend.pos = "topright",
  legend.title.txt = "My first map : cod_sectir",
  legend.values.rnd = -2
)
```



3 Map and Calculus of area indicators: CARTE NBD BTT

Here
- Calculation by zone: average and median of the daily travel time on individuals in the area and the number of trips and total number of individuals in the area
- Reduction of the DF to lighten the execution
- Merging of the DF with the GIS DF (which contains points coordinates and polygons)

```
DF <- allgreI[, c("nbd", "btt", "tir", "zoneres.y", "id_pers","D12_sum")]
# Calcul base individus des indicateurs individuels
DF_Indiv <- na.omit(DF)
addmargins(table(allgreI$zoneres.y))

DF_Indiv$BTT_moy <- ave(DF_Indiv$btt,DF_Indiv$tir, FUN = mean)
DF_Indiv$BTT_med <- ave(na.omit(DF_Indiv$btt),DF_Indiv$tir, FUN = median)
DF_Indiv$nbd_med <- ave(DF_Indiv$nbd,DF_Indiv$tir, FUN = median)
DF_Indiv$nbd_moy <- ave(DF_Indiv$nbd,DF_Indiv$tir, FUN = mean)
DF_Indiv$Maxind <- ave(DF_Indiv$id_pers,DF_Indiv$tir, FUN = max)
DF_Indiv$TD12_moy <- ave(DF_Indiv$D12_sum,DF_Indiv$tir, FUN = mean)
DF_Indiv$TD12_med <- ave(na.omit(DF_Indiv$D12_sum),DF_Indiv$tir, FUN = median)

# Réduction à la base Zone (avec le max(ind) par TIR)
DF_Zone <- subset(DF_Indiv, DF_Indiv$id_pers == DF_Indiv$Maxind, select =c(zoneres.y,BTT_med, BTT_moy, n
```

3.0.1 Merging with newESRI and data

```
# library(sp) # the trick is that this package must be loaded!
# MERGING with option duplicateGEOM=T to get as many polygons as the number of observationsin the data
DF_Zone$cod_sectir <- DF_Zone$tir
newESRI_DF <- merge(newESRI, DF_Zone, by = "cod_sectir", duplicateGeoms = TRUE)
```

3.0.2 Projecting your geographic coordinates to the equirectangular projection

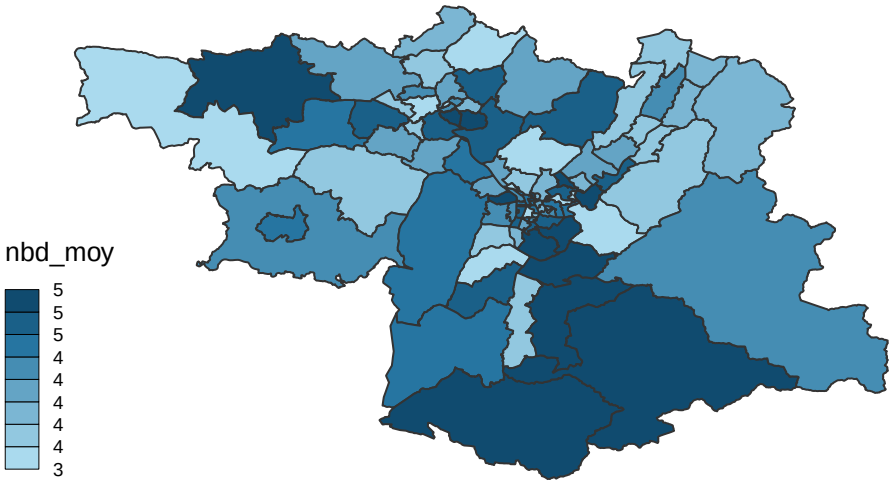
After merging the data and the map, geographic coordinates have to be projected.

https://r-spatial.org/r/2020/06/17/s2.html

```
newESRI_DF = st_transform(newESRI_DF, "+proj=eqc")
```

4 Colored mapping

```
choroLayer(spdf = newESRI_DF, var = "nbd_moy")
title("Mapping of the area number - not really useful")
```



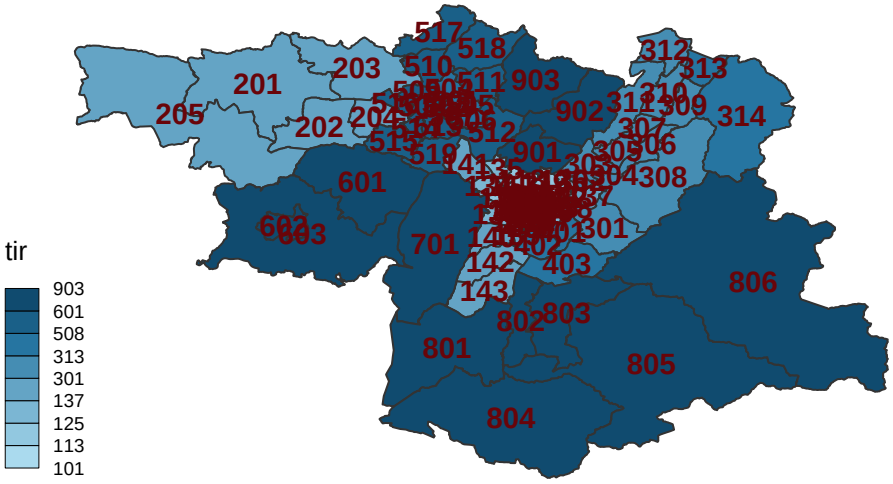

```
table(newESRI_DF$tir)
```

```
##
## 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 141 142 143 201 202 203 204 205 301 302 303 304 305 306 307 308 309 310 311 312
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 313 314 401 402 403 501 502 503 504 505 506 507 508 509 510
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  [ reached 'max' / getOption("max.print") -- omitted 22 entries ]
```

5 Map with labels

It may need to install `lwgeom`

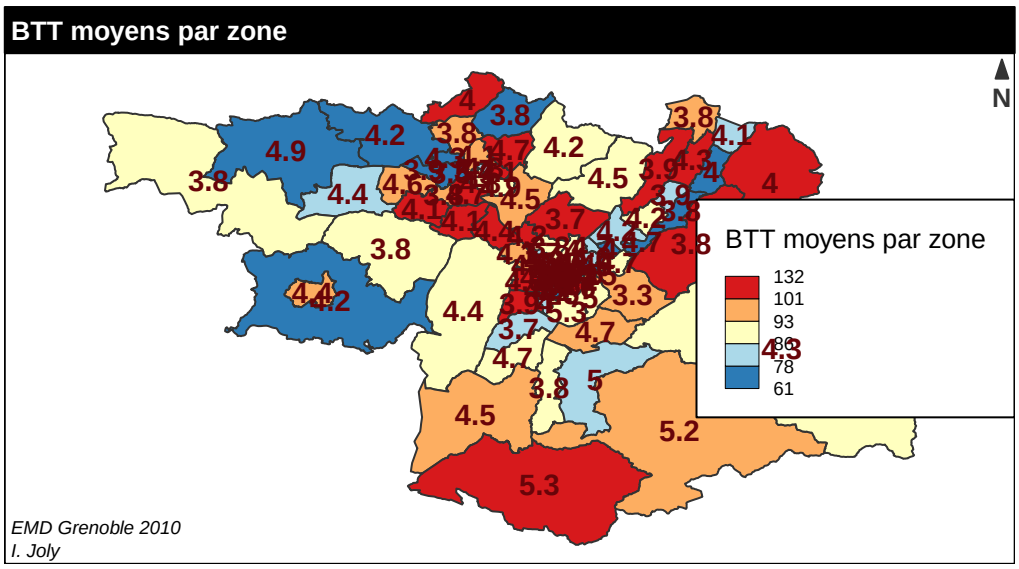
```
choroLayer(spdf = newESRI_DF,var = "tir")
# Label plot of the Mean
labelLayer(spdf = newESRI_DF, txt = "tir", col = "#690409", cex = 0.9, font = 2)
```



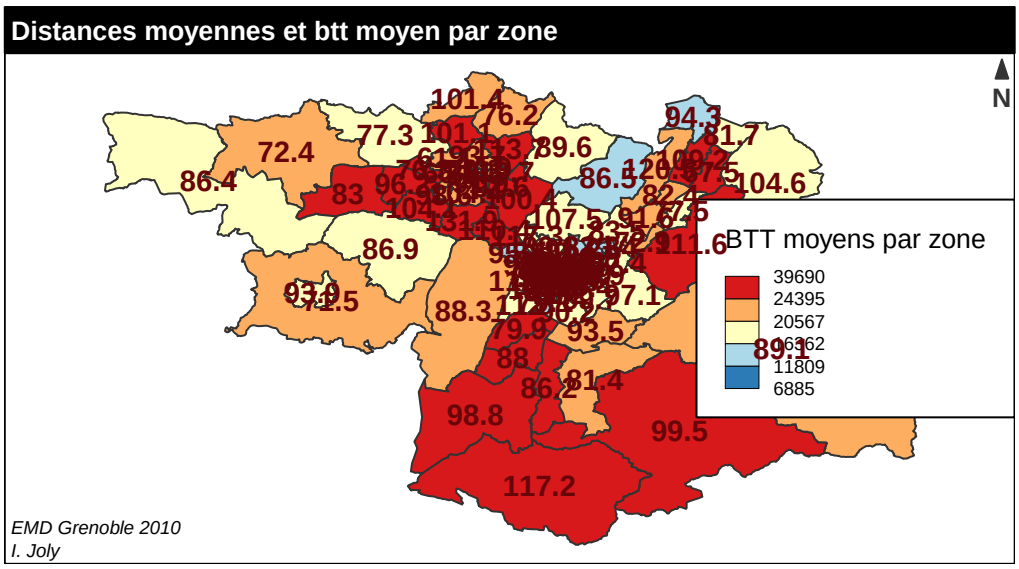
6 Complete map of daily travel times

The script may need to be run out of MARKDOWN

```
Nclasse <- 5
########## Carte colorée
choroLayer(spdf = newESRI_DF,
           #df = DF_Zone,
           spdfid = "cod_sectir", # id in map data
           dfid = "tir",  # id in dataframe
           var = "BTT_moy",
           nclass = Nclasse,
           col = rev(brewer.pal(n = Nclasse, "RdYlBu")),
           legend.pos = "right", legend.frame = TRUE,
           legend.title.txt = "BTT moyens par zone")
########### Ajout de label : nbd par zone
# Label creation
newESRI_DF$lab <- paste(round(newESRI_DF$nbd_moy,1), sep ="")
# Label plot of the Mean
labelLayer(spdf = newESRI_DF,  txt = "lab", col = "#690409", cex = 0.9, font = 2)
####### Ajout d'éléments de présentations : Layout plot
layoutLayer(title = "BTT moyens par zone",
            sources = "EMD Grenoble 2010",
            author = "I. Joly",
            scale = 0,
            north = T,
            frame = TRUE,
            col = "black",
            coltitle = "white")
```




```
Nclasse <- 5
########## Carte colorée
choroLayer(spdf = newESRI_DF,
           #df = DF_Zone,
           spdfid = "cod_sectir", # id in map data
           dfid = "tir",  # id in dataframe
           var = "TD12_moy",
           nclass = Nclasse,
           col = rev(brewer.pal(n = Nclasse, "RdYlBu")),
           legend.pos = "right", legend.frame = TRUE,
           legend.title.txt = "BTT moyens par zone")
############ Ajout de label : nbd par zone
# Label creation
newESRI_DF$lab <- paste(round(newESRI_DF$BTT_moy,1), sep ="")
# Label plot of the Mean
labelLayer(spdf = newESRI_DF,  txt = "lab", col = "#690409", cex = 0.9, font = 2)
####### Ajout d'éléments de présentations : Layout plot
layoutLayer(title = "Distances moyennes et btt moyen par zone",
            sources = "EMD Grenoble 2010",
            author = "I. Joly",
            scale = 0,
            north = T,
            frame = TRUE,
            col = "black",
            coltitle = "white")
```


Distances moyennes et btt moyen par zone
N
BTT moyens par zone
39690
24395
20567
16362
11809
6885
101.4
76.2
94.3
81.7
77.3
72.4
89.6
86.4
86.5
104.6
83
107.5
111.6
86.9
93.0
72.5
97.1
88.3
79.8
93.5
89.1
88
81.4
86.2
98.8
99.5
117.2
EMD Grenoble 2010
I. Joly

7 References

Chang, Winston. 2013. *R Graphics Cookbook*. O'Reilly Media, Inc.

Harrell, F. E. 2013. *Regression Modeling Strategies: With Applications to Linear Models, Logistic Regression, and Survival Analysis*. Springer Series in Statistics. Springer New York. https://books.google.fr/books?id=7D0mBQAAQBAJ.

Munzner, Tamara. 2015. *Visualization Analysis and Design*. CRC Press, Routledge.

Xie, Yihui. 2015. *Dynamic Documents with R and Knitr*. 2nd ed. Chapman; Hall/CRC. https://yihui.name/knitr/.

Xie, Yihui. 2016. *Bookdown: Authoring Books and Technical Documents with R Markdown*. Chapman; Hall/CRC. https://github.com/rstudio/bookdown.

- Spatial data with `sf`: https://r-spatial.github.io/sf/

7.1 Sources

8 Appendix

Here you can add your material